```
#include <iostream>
Using namespace std;

Struct Node {
   Int data;
   Node *left, *right;
   Bool lthread, rthread;
};

Node* createNode(int value) {
   Node* temp = new Node;
   Temp->data = value;
   Temp->left = temp->right = nullptr;
   Temp->lthread = temp->rthread = true;
   Return temp;
}

Node* insert(Node* root, int key) {
   Node* ptr = root;
   Node* parent = nullptr;

   While (ptr != nullptr) {
      If (key == ptr->data) {
```

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        Cout << “Duplicate value not allowed\n”;
        Return root;
    }

    Parent = ptr;

    If (key < ptr->data) {
        If (!ptr->lthread)
            Ptr = ptr->left;
        Else
            Break;
    } else {
        If (!ptr->rthread)
            Ptr = ptr->right;
        Else
            Break;
    }
}

Node* temp = createNode(key);

If (parent == nullptr) {
    Root = temp;
```

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    }
    Else if (key < parent->data) {
        Temp->left = parent->left;
        Temp->right = parent;
        Parent->lthread = false;
        Parent->left = temp;
    }
    Else {
        Temp->right = parent->right;
        Temp->left = parent;
        Parent->rthread = false;
        Parent->right = temp;
    }

    Return root;
}

Node* inorderPredecessor(Node* ptr) {
    If (ptr->lthread)
        Return ptr->left;

    Ptr = ptr->left;
```

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    While (!ptr->rthread)
        Ptr = ptr->right;

    Return ptr;
}

Node* inorderSuccessor(Node* ptr) {
    If (ptr->rthread)
        Return ptr->right;

    Ptr = ptr->right;

    While (!ptr->lthread)
        Ptr = ptr->left;

    Return ptr;
}

Int main() {
    Node* root = nullptr;
    Int n, value, key;

    Cout << “Enter number of nodes: “;
```

```
Cin >> n;

Cout << “Enter node values: “;
For (int I = 0; I < n; i++) {
    Cin >> value;
    Root = insert(root, value);
}

Cout << “Enter node to find predecessor and successor: “;
Cin >> key;

Node* ptr = root;

While (ptr != nullptr && ptr->data != key) {
    If (key < ptr->data) {
        If (ptr->lthread)
            Ptr = nullptr;
        Else
            Ptr = ptr->left;
    } else {
        If (ptr->rthread)
            Ptr = nullptr;
        Else
```

```
            Ptr = ptr->right;
        }
}

If (ptr == nullptr) {
    Cout << “Node not found\n”;
    Return 0;
}

Node* pred = inorderPredecessor(ptr);
Node* succ = inorderSuccessor(ptr);

If (pred != nullptr)
    Cout << “Inorder Predecessor: “ << pred->data << endl;
Else
    Cout << “Inorder Predecessor does not exist\n”;

If (succ != nullptr)
    Cout << “Inorder Successor: “ << succ->data << endl;
Else
    Cout << “Inorder Successor does not exist\n”;

Return 0;
```

}